

POSITIVITY OF STRETCHED LITTLEWOOD–RICHARDSON COEFFICIENTS FOR PARTITIONS OF LENGTH AT MOST FOUR

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ABSTRACT. For partitions λ, μ, ν the Littlewood–Richardson coefficient $c_{\lambda\mu}^\nu$ stretches to a function $P(t) = c_{t\lambda, t\mu}^{t\nu}$ which, by a theorem of Derksen and Weyman, is a polynomial in t . King, Tollu and Toumazet conjectured that P has no negative coefficient. The conjecture is known only for $c_{\lambda\mu}^\nu \leq 2$ and is otherwise open; it is recorded as an unsolved problem. We prove it for all triples whose parts number at most four. The proof is structural: a period-one dilation transfers the Berline–Vergne local Ehrhart formula from an integral dilate back to the rational hive polytope, reducing the statement to the positivity of the Berline–Vergne weight of every two-dimensional transverse cone spanned by rank-four rhombus normals, which we verify exactly (minimum weight $1/9$). Vertex integrality and unimodularity of vertex cones are not used, and indeed fail already in rank four. The same local method yields two rank-uniform consequences: for every full-dimensional hive polytope of any rank the top four Ehrhart coefficients are positive, and every full-dimensional five-part hive has all coefficients positive except possibly the linear one. We close by reducing the general conjecture to a single hive-specific effectivity statement and recording the exact obstructions that rule out the standard shortcuts. All finite verifications are accompanied by independent replay scripts.

1. INTRODUCTION

Let λ, μ, ν be partitions with $|\lambda| + |\mu| = |\nu|$ and let $c_{\lambda\mu}^\nu$ be the Littlewood–Richardson (LR) coefficient, the multiplicity of the Schur function s_ν in $s_\lambda s_\mu$. For a positive integer t write $t\lambda = (t\lambda_1, t\lambda_2, \dots)$. Derksen and Weyman [3] proved that

$$P_{\lambda\mu}^\nu(t) := c_{t\lambda, t\mu}^{t\nu}$$

agrees, for all $t \geq 1$, with a polynomial in t ; a short proof was later given in [15]. King, Tollu and Toumazet [8, 9] studied these *stretched* coefficients in detail and conjectured a strong positivity property.

Conjecture 1 (King–Tollu–Toumazet). Every coefficient of $P_{\lambda\mu}^\nu(t)$, in the monomial basis $1, t, t^2, \dots$, is nonnegative.

Conjecture 1 is proved for $c_{\lambda\mu}^\nu = 1$ (where $P \equiv 1$, by the Fulton conjecture of Knutson–Tao–Woodward [7]) and for $c_{\lambda\mu}^\nu = 2$ (where $P(t) = t + 1$, by Ikenmeyer [5] and geometrically by Sherman [14]). Beyond these it is open, and it is listed as an unsolved problem in the FrontierMath collection [4]. Coquereaux and Zuber [2] record the special case of the linear coefficient in the four-part ($SU(4)$) situation and leave its nonnegativity as an unproved (“alleged”) assertion. Extensive computation has produced no counterexample.

We settle the four-part case.

Theorem 2. *Let λ, μ, ν be partitions of length at most four with $|\lambda| + |\mu| = |\nu|$. Then every coefficient of $P_{\lambda\mu}^\nu(t)$ is nonnegative. If $c_{\lambda\mu}^\nu > 0$ and the associated hive polytope is three-dimensional, all four coefficients are strictly positive.*

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This is a computer-assisted proof; all finite verifications are reproduced by the scripts described in Section 8.

The proof is geometric. By the hive theorem of Knutson and Tao [6] the coefficient $c_{\lambda\mu}^\nu$ counts the lattice points of the *hive polytope* $H = H(\lambda, \mu, \nu)$, a rational polytope in the space of interior hive entries; stretching dilates H , so $P_{\lambda\mu}^\nu$ is the Ehrhart polynomial of H and its degree is $\dim H$. For length at most four, $\dim H \leq 3$. Ehrhart positivity is automatic in dimensions ≤ 2 , so the whole difficulty is the linear coefficient of a three-dimensional hive.

Two features make this delicate. First, hive polytopes for four parts are lattice polytopes, but this is a nontrivial input; we avoid it entirely by a dilation argument (Section 3), which also makes the method work verbatim in the rational case that occurs for five or more parts. Second, Ehrhart positivity is *false* for general lattice polytopes—the Reeve tetrahedra have negative linear coefficient—so any proof must use the special geometry of hives. Ours uses the Berline–Vergne local Ehrhart formula [1, 13]: the linear coefficient is a sum, over the edges, of the lattice length of the edge times a weight depending only on the edge’s transverse cone. Because every rank-four rhombus inequality has a primitive normal with entries in $\{0, \pm 1\}$, there are finitely many transverse cones, and we check that every one of their Berline–Vergne weights is positive. We emphasise that this is a statement about *edges*: vertex cones of four-part hives need not be unimodular (Section 5.1), but this does not affect an edge-local argument.

The same local computation, carried out uniformly in the number of parts, gives the following.

Theorem 3. *For every full-dimensional hive polytope of intrinsic dimension D (any number of parts), the coefficients of $t^D, t^{D-1}, t^{D-2}, t^{D-3}$ in its Ehrhart polynomial are positive.*

Theorem 4. *For every full-dimensional five-part hive polytope (necessarily of dimension six) all Ehrhart coefficients except possibly the linear one are positive.*

Theorem 3 is a statement about *full-dimensional* hives; for such a hive of dimension $D \leq 4$ it gives all coefficients, but the only full-dimensional hives with $D \leq 4$ are the four-part ones, so its new content is the top four coefficients of full-dimensional hives of dimension $D \geq 5$. Together with Theorem 2 it leaves, for five parts, only the linear coefficient of a full-dimensional hive undetermined (Theorem 4), together with the non-full-dimensional five-part hives of intrinsic dimension at least three, whose face transverse cones are projections of N_5 and are not covered by the atlas.

Finally, in Section 7 we make precise the obstruction to a general proof. The local formula reduces Conjecture 1 to a hive-specific “effectivity” statement (HTE) about Berline–Vergne/Todd weights of closed flat-rhombus coarsenings; we prove that termwise (simplex-cone) positivity is *false* at codimension four, so a general proof must control the closed coarsenings globally, and we record six standard rank-uniform shortcuts together with the exact examples that defeat each. These do not diminish the four-part theorem, which is complete.

The finite verifications underlying Theorems 2–4 are exact (integer and rational arithmetic throughout) and are reproduced by independent scripts described in Section 8; the Littlewood–Richardson counts were cross-checked against two independent tableau/hive enumerators and against published values [12].

2. THE HIVE MODEL AND THE LOCAL EHRHART FORMULA

We recall the hive description of LR coefficients [6] and fix coordinates for the four-part case.

A *hive* of size r is a real function on the triangular array $\{(i, j) : 0 \leq j \leq i \leq r\}$ that is concave across each of the three families of unit rhombi. With border values $h(i, 0) = \lambda_1 + \cdots + \lambda_i$, $h(i, i) = \nu_1 + \cdots + \nu_i$, and $h(r, j) = |\lambda| + \mu_1 + \cdots + \mu_j$, the integer hives are in bijection with the LR tableaux counted by $c_{\lambda\mu}^\nu$. The interior entries are free subject to the rhombus inequalities, so the hive polytope $H(\lambda, \mu, \nu) \subset \mathbb{R}^D$, $D = \binom{r-1}{2}$, is cut out by a fixed integer matrix A_r (depending only on r) applied to the interior entries, with a right-hand side that is an integral, homogeneous, linear function of (λ, μ, ν) . Consequently

$$H(t\lambda, t\mu, t\nu) = tH(\lambda, \mu, \nu), \quad L_H(t) := \#(tH \cap \mathbb{Z}^D) = P_{\lambda\mu}^\nu(t),$$

so $P_{\lambda\mu}^\nu$ is the Ehrhart polynomial of H and $\deg P_{\lambda\mu}^\nu = \dim H$. For $r = 4$ we have $D = 3$ and coordinates (h_{11}, h_{12}, h_{21}) ; eliminating the border, the 18 rhombus rows reduce to 15 distinct primitive normals, all with entries in $\{0, \pm 1\}$ (Section 4).

We use the Berline–Vergne local Euler–Maclaurin formula [1] (see also [13] for the presentation we adopt). For a rational polytope $Q \subset \mathbb{R}^D$ and each k , the coefficient of t^k in $L_Q(t)$ equals

$$(1) \quad [t^k] L_Q(t) = \sum_{\dim F=k} \alpha(N(Q, F)) \operatorname{vol}_{\mathbb{Z}}(F),$$

where the sum runs over the k -dimensional faces F , $\operatorname{vol}_{\mathbb{Z}}(F)$ is the relative lattice volume of F , and α is a weight depending only on the transverse cone $N(Q, F)$ of Q along F , and invariant under lattice translation of that cone. The leading term ($k = D$) is the normalized volume and the codimension-one term ($k = D - 1$) is half the normalized surface area; both are manifestly positive.

3. PERIOD-ONE DILATION

The next lemma removes any need to know that a hive polytope is a lattice polytope, and lets us apply the lattice formula (1) to the rational polytopes that occur for five or more parts.

Lemma 5. *Fix r and let N_r be the (finite) set of primitive rhombus normals. Suppose that for every lattice polytope $P \subset \mathbb{R}^D$ whose facet normals lie in N_r one has $[t^k] L_P(t) \geq 0$. Then $[t^k] P_{\lambda\mu}^\nu(t) \geq 0$ for all partitions of length at most r .*

Proof. Let $H = H(\lambda, \mu, \nu) = \{x : A_r x \leq b\}$ with b integral, and let q be the least common multiple of the denominators of the vertices of H . Then $qH = \{x : A_r x \leq qb\}$ has the same normal fan as H —so its facet normals lie in N_r —and its vertices are q times those of H , hence integral; thus qH is a lattice polytope. Since $L_{qH}(t) = \#(t qH \cap \mathbb{Z}^D) = L_H(qt)$, we have $[t^k] L_{qH} = q^k [t^k] L_H$ with $q > 0$, so the two coefficients have the same sign. Applying the hypothesis to qH gives $[t^k] L_H \geq 0$. $\square$

By Lemma 5 it suffices, for each fixed r , to prove Ehrhart coefficient positivity for lattice polytopes with normals in N_r ; the polynomial $P_{\lambda\mu}^\nu$ inherits it. In particular the four-part theorem does not depend on the integrality of four-part hives.

4. THE RANK-FOUR NORMAL ATLAS

In coordinates (h_{11}, h_{12}, h_{21}) the primitive rank-four rhombus normals are the 15 vectors

$$\pm e_1, \pm e_2, \pm e_3, \pm(e_1 - e_2), \pm(e_1 - e_3), \pm(e_2 - e_3),$$

$$(1, -1, -1), (-1, 1, -1), (-1, -1, 1),$$

each with entries in $\{0, \pm 1\}$. There are $\binom{15}{2} - 6 = 99$ nonparallel pairs. Twelve of the fifteen normals are the type- A_3 “alcoved” directions; the three remaining “odd” rows are what make the atlas fail to be totally unimodular (over $\binom{15}{3} = 455$ triples of rows the determinant multiset is $\{|\det| = 0:146, 1:272, 2:36, 4:1\}$).

5. PROOF OF THEOREM 2

By Lemma 5 we may assume H is a nonempty lattice polytope in $\mathbb{R}^3$ with normals in N_4 ; write $L_H(t) = a_3 t^3 + a_2 t^2 + a_1 t + 1$. (If $c_{\lambda\mu}^\nu = 0$ then $P \equiv 0$ by saturation [6], and the statement is vacuous.) The leading coefficient a_3 is the normalized volume and a_2 is half the normalized surface area; both are positive, and if $\dim H \leq 2$ the polynomial is a point, segment, or polygon Ehrhart polynomial, which is coefficientwise positive. It remains to bound a_1 for $\dim H = 3$.

Apply (1) with $k = 1$: since a three-dimensional polytope has each edge E as a ridge lying in exactly two facets, the transverse cone $N(H, E)$ is the two-dimensional cone spanned by the two facet normals meeting along E , both in N_4 . Hence

$$a_1 = \sum_{\text{edges } E} \alpha(N(H, E)) \ell_{\mathbb{Z}}(E),$$

with $\ell_{\mathbb{Z}}(E) \geq 0$ the lattice length of E .

Proposition 6. *For every ordered pair of nonparallel normals in N_4 , the Berline–Vergne weight α of the corresponding pointed two-dimensional transverse cone is positive; over the 99 pairs its minimum value is $1/9$.*

Proposition 6 is a finite exact computation (Section 8); the weight of a saturated normal basis with Gram matrix $\begin{pmatrix} A & C \\ C & B \end{pmatrix}$ equals $\frac{1}{4} - \frac{C}{12}(\frac{1}{A} + \frac{1}{B})$, and evaluating this over the atlas gives values in $\{1/9, 5/18, 1/4\}$, all positive. Granting it,

$$a_1 \geq \frac{1}{9} \sum_{\text{edges } E} \ell_{\mathbb{Z}}(E) > 0,$$

so all four coefficients of L_H are positive. Undoing the dilation (Lemma 5) proves Theorem 2. $\square$

5.1. Nonsimple vertices do not obstruct the argument. The hypotheses use only that a_1 is edge-local. For definiteness we note that four-part hives really do have nonunimodular vertex cones: at $\lambda = \mu = (12, 8, 4)$, $\nu = (18, 14, 10, 6)$ the vertex $(26, 32, 38)$ is simple with primitive tangent rays $(0, 1, 1), (1, 0, 1), (1, 1, 0)$, whose determinant is 2 (and the polytope has $c'_{\lambda\mu} = 50$, all vertices integral, $L_H(t) = 1 + 7t + 18t^2 + 24t^3$). This has no bearing on (1) for $k = 1$: each edge still lies in exactly two facets, and any further rhombus rows tight along an edge are redundant on, or interior to, its two-dimensional transverse cone.

6. RANK-UNIFORM COEFFICIENTS AND FIVE PARTS

Formula (1) localizes each coefficient on faces of a fixed codimension, whose transverse cones, for hives of any rank, are generated by the fixed normal set N_r . Enumerating the possible closed transverse cones and evaluating α exactly gives Theorem 3: the volume and surface terms handle codimensions 0, 1, and exact rank-uniform positivity of α on every closed codimension-two and codimension-three hive transverse cone handles t^{D-2} and t^{D-3} .

For Theorem 4 ($r = 5$, $D = 6$) one further coefficient, the quadratic, is controlled at codimension four. Here termwise positivity already fails—an individual saturated four-normal cell can have weight $-349/28800$ —but the exact slack closure forces such a cell either to raise the normal rank to at least five or to embed in one of finitely many pointed rank-preserving closed supersets, and every one of those full cones has positive weight (minimum $739/86400$). Enumerating all $\binom{27}{4} = 17550$ four-normal tuples and their closures (903 nonsaturated tuples with minimum weight $1/14400$; 132 negative saturated cells, each repaired) yields positivity of $t^6, \dots, t^2$, leaving only the linear coefficient. The full enumeration and its independent replay are in the codimension-four report.

Theorem 4 concerns full-dimensional five-part hives. The lower-dimensional five-part hives are not reduced to four parts by Horn-facet factorization: the triple $\lambda = (4, 3, 3, 1)$, $\mu = (4, 2, 1, 1)$, $\nu = (6, 5, 4, 2, 2)$ has $\dim H = 3$ while all 142 essential Horn inequalities are strict.

7. THE GENERAL FRONTIER

For a period-one hive polytope the local formula writes every Ehrhart coefficient as a pairing of nonnegative face volumes with Berline–Vergne/Todd weights on transverse cones. Thus Conjecture 1 follows from a single hive-specific effectivity statement: for every rank, codimension q ,

and 2-connected primitive-interior closed flat-rhombus coarsening Σ with weight vector a_q and Minkowski boundary map ∂_q ,

$$(HTE) \quad \exists y: \quad a_q + \partial_q^\top y \geq 0.$$

By rational Farkas duality (HTE) is equivalent to $\langle a_q, w \rangle \geq 0$ for every nonnegative balanced realizable face weight w ; pairing it with the (positive) face-volume weight of any hive polytope makes every coefficient nonnegative. Horn factorization [9, 10] disposes of separating boundary coarsenings, so the load-bearing case is the 2-connected primitive interior. We have neither a proof of (HTE) nor a negative realizable balanced weight refuting it.

On the disproof side, one transfer route deserves record because it is *valid* and still fails. Realizing three disjoint horizontal strips as a skew shape identifies the lattice points of any $3 \times N$ transportation polytope with Littlewood–Richardson tableaux of a fixed triple, compatibly with dilation: for example, the polytope with row margins $(3, 3, 3)$ and column margins $(2, 1^7)$ has $L_T(n) = c_{n\lambda, n\mu}^{n\nu}$ for $\lambda = (9, 6, 3)$, $\mu = (9, 9, 7, 6, 5, 4, 3, 2, 1)$, $\nu = (15, 12, 9, 7, 6, 5, 4, 3, 2, 1)$ (both counts equal 1050 at $n = 1$). Hence a 3×8 transportation polytope with a negative Ehrhart coefficient would refute Conjecture 1 outright. None exists in the relevant regime: over the full dimension-14, codegree-3 margin family the minimum linear coefficient is $2157/280 > 0$ (exhaustive for total margin ≤ 19 , with saturation in every unbounded direction, and with the closed form $\frac{3a}{2}(1 + H_{2a-1} - H_a) > 0$ for the symmetric unit-column subfamily); moreover codegree 3 forces $L_T(1) \geq 1050$, so the known negative order polytope of Liu and Tsuchiya [11], which has 255 lattice points, is not realizable in this shape. The route evades the positivity theorems for skew Gelfand–Tsetlin polytopes (its content constraints are global sums, not markings) and dies only on these measured facts.

Termwise (simplex-cone) positivity is genuinely false—the $-349/28800$ cell above—so a general proof cannot proceed cell by cell; the finite success at codimension four does not extend to an unbounded certificate cascade. We record, finally, that the standard rank-uniform shortcuts are individually obstructed by exact examples: generic type- A Todd/cocircuit balancing (a nonnegative A_4 Dahmen–Micchelli polynomial maps to $N^4 + 4N^3 + 3N^2 - 2N - \frac{6}{5}$); network-flow equivalence (simple hive tangent cones have determinant two); generic Cohen–Macaulay/Koszul Hilbert-ring structure (the Q_{20} Hibi ring is Gorenstein, Koszul, with negative a -invariant, yet its Hilbert polynomial has linear coefficient $-168011/330$); matroidal/secondary-fan Todd effectivity (a genuine hive face carries a $U_{2,4}$ tight-normal restriction); positive tableau/Kostka recursions (they require signs, or stop on primitive triples); and rank-by-rank Hurwitz stability (an unbounded hierarchy). Details and the supporting computations are in the accompanying reports.

8. REPRODUCIBILITY

Every finite claim above is verified by exact arithmetic and reproduced by scripts distributed with this paper.

- Two independent Littlewood–Richardson counters (a hive lattice-point enumerator and a lattice-word LR-tableau counter), cross-checked against each other on all triples with $|\nu| \leq 8$, against $c = 1, 2$ stretched values, and against [12].
- The rank-four atlas of 15 normals and 99 pairs, and Proposition 6 (all Berline–Vergne weights, minimum $1/9$).
- An independent Ehrhart interpolation that recomputes $L_H(0), \dots, L_H(D+2)$ and confirms the interpolated polynomial at two held-out points, checked against both LR counters; and a verification of $a_1 = \sum_E \alpha \ell_{\mathbb{Z}}(E)$ against lattice-count a_1 on more than 10^4 four-part hives that are not among the atlas basis polytopes.
- The rank-uniform codimension-two and codimension-three enumerations, and the full codimension-four five-part enumeration with the closure repairs.

The dilation identity of Lemma 5 and the nonunimodular vertex of Section 5.1 are likewise reproduced exactly.

The complete reproducibility bundle—the two Littlewood–Richardson counters, the rank-four atlas and codimension-two/three enumerations, the five-part codimension-four certificate and its independent checker, and the exact replay scripts for every displayed value—is available at <https://github.com/AlperTheKing/ktt-positivity>.

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